

SF1811 Optimization

Lecture 15: Global optimality, Lagrangian relaxation, and outlook

Chapters 21–22 and course summary

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Global optimality, Lagrangian relaxation, and outlook

Main questions:

How can local optimality conditions become global certificates?

How can problem structure be exploited in larger optimization problems?

Today:

- KKT conditions and global optimality.
- Lagrangian relaxation and dual bounds.
- Separable problems and decomposition.
- Solutions and optimality certificates.
- Outlook beyond this course.

The constrained optimization problem

Consider

$$(P) \qquad \begin{array}{ll} \text{minimize} & f(x) \\ \text{subject to} & h(x) = 0, \\ & g(x) \leq 0, \\ & x \in X. \end{array}$$

The Lagrangian is

$$L(x, u, y) = f(x) + u^{\mathsf{T}} h(x) + y^{\mathsf{T}} g(x), \qquad y \geq 0.$$

From L13, the KKT conditions include

$$\nabla_x L(\hat{x}, \hat{u}, \hat{y}) = 0,$$

primal feasibility, dual feasibility, and complementarity.

For a general nonlinear problem, these are local first-order conditions.

Stationarity need not imply global minimization of the Lagrangian.

From stationarity to global minimization

KKT stationarity states that

$$\nabla_x L(\hat{x}, \hat{u}, \hat{y}) = 0.$$

The stronger global condition is

$$\boxed{L(\hat{x}, \hat{u}, \hat{y}) = \min_{x \in X} L(x, \hat{u}, \hat{y}).}$$

The distinction is important:

- stationarity uses local derivative information at $\hat{x}$;
- minimization of the Lagrangian compares $\hat{x}$ with every $x \in X$.

If $L(\cdot, \hat{u}, \hat{y})$ is convex, stationarity is sufficient for global minimization of the Lagrangian.

Convexity turns the local stationarity condition into a global one.

Global optimality conditions

Global optimality conditions

A point $(\hat{x}, \hat{u}, \hat{y})$ satisfies the global optimality conditions if:

1. $L(\hat{x}, \hat{u}, \hat{y}) = \min_{x \in X} L(x, \hat{u}, \hat{y})$. (Global minimization of the Lagrangian)
2. $h(\hat{x}) = 0$, $g(\hat{x}) \leq 0$, $\hat{x} \in X$. (Primal feasibility)
3. $\hat{y} \geq 0$. (Dual feasibility)
4. $\hat{y}^{\mathsf{T}} g(\hat{x}) = 0$. (Complementarity)

Theorem

If $(\hat{x}, \hat{u}, \hat{y})$ satisfies the global optimality conditions, then $\hat{x}$ is a global minimizer of (P).

Why do the global conditions certify optimality?

Let x be any feasible point of (P). Since $\hat{y} \geq 0$ and $g(x) \leq 0$,

$$L(x, \hat{u}, \hat{y}) = f(x) + \hat{u}^{\mathsf{T}} h(x) + \hat{y}^{\mathsf{T}} g(x) \leq f(x).$$

Global minimization of the Lagrangian gives

$$L(\hat{x}, \hat{u}, \hat{y}) \leq L(x, \hat{u}, \hat{y}).$$

Primal feasibility and complementarity give

$$L(\hat{x}, \hat{u}, \hat{y}) = f(\hat{x}).$$

Therefore,

$$\boxed{f(\hat{x}) \leq L(x, \hat{u}, \hat{y}) \leq f(x).}$$

Thus $\hat{x}$ is a global minimizer of (P).

The convex case

Assume that

- f is convex;
- each component of g is convex;
- each component of h is affine;
- X is convex.

For every u and every $y \geq 0$, the function

$$x \longmapsto L(x, u, y)$$

is convex on X.

If $(\hat{x}, \hat{u}, \hat{y})$ satisfies the KKT conditions, then

$$\nabla_x L(\hat{x}, \hat{u}, \hat{y}) = 0$$

implies

$$L(\hat{x}, \hat{u}, \hat{y}) = \min_{x \in X} L(x, \hat{u}, \hat{y}).$$

Hence the global optimality conditions hold.

Lagrangian relaxation

For fixed multipliers u and $y \geq 0$, relax the constraints $h(x) = 0$ and $g(x) \leq 0$ and solve

$$(P_R(u, y)) \qquad \underset{x \in X}{\text{minimize}} \quad L(x, u, y).$$

Its optimal value defines the dual function:

$$\boxed{\varphi(u, y) = \min_{x \in X} L(x, u, y).}$$

For every feasible x of (P),

$$L(x, u, y) = f(x) + u^{\mathsf{T}} h(x) + y^{\mathsf{T}} g(x) \leq f(x).$$

Therefore,

$$\varphi(u, y) \leq \text{optval}(P).$$

Every feasible choice of multipliers gives a lower bound.

The Lagrangian dual problem

The Lagrangian dual problem searches for the best lower bound:

$$(D) \qquad \begin{array}{ll} \text{maximize} & \varphi(u, y) \\ \text{subject to} & y \geq 0. \end{array}$$

Weak duality gives

$$\text{optval}(D) \leq \text{optval}(P).$$

If

$$\varphi(\hat{u}, \hat{y}) = f(\hat{x})$$

for a primal feasible $\hat{x}$ and dual feasible $(\hat{u}, \hat{y})$, then

$$\hat{x} \text{ is primal optimal}, \qquad (\hat{u}, \hat{y}) \text{ is dual optimal}.$$

Equal primal and dual values provide an optimality certificate.

Looking back to linear programming

In L5, the LP dual arose from simplex optimality and strong duality.

In L6, the same dual problem arose from Lagrangian relaxation.

For linear programming,

$$\text{optval}(D) = \text{optval}(P)$$

whenever an optimal solution exists.

The primal and dual optimal solutions satisfy complementarity.

The same ideas now appear in a broader setting:

- multipliers produce lower bounds;
- complementarity identifies active constraints;
- convexity can make first-order conditions globally sufficient;
- primal and dual solutions together certify optimality.

Why solve a Lagrangian relaxation?

The relaxed problem

$$\min_{x \in X} L(x, u, y)$$

is useful when it is easier than the original constrained problem.

This may happen because:

- the relaxed problem has an analytical solution;
- the variables separate into independent groups;
- a difficult coupling constraint has been removed;
- the remaining problem has network or dynamic-programming structure;
- the resulting subproblems can be solved in parallel.

The goal is not merely to remove constraints. The goal is to remove the constraints that prevent useful structure from being exploited.

A good relaxation preserves easy structure and removes difficult coupling.

A separable problem with a coupling constraint

Consider

$$\begin{aligned} \text{minimize} \quad & \sum_{j=1}^{N} f_j(x_j) \\ \text{subject to} \quad & \sum_{j=1}^{N} A_j x_j = b, \\ & x_j \in X_j, \quad j = 1, \ldots, N. \end{aligned}$$

The objective and the simple sets X_j are separable, but the equality constraint couples all variable blocks.

Relax the coupling constraint with a multiplier u:

$$L(x, u) = \sum_{j=1}^{N} f_j(x_j) + u^{\mathsf{T}} \left(b - \sum_{j=1}^{N} A_j x_j \right).$$

Rearranging gives $L(x, u) = b^{\mathsf{T}} u + \sum_{j=1}^{N} \left(f_j(x_j) - u^{\mathsf{T}} A_j x_j \right)$.

The relaxation separates

The dual function is

$$\begin{aligned}\varphi(u) &= \min_{x_j \in X_j,\ j=1,\ldots,N} L(x,u) \\ &= b^{\mathsf{T}}u + \sum_{j=1}^{N} \min_{x_j \in X_j} \left(f_j(x_j) - u^{\mathsf{T}} A_j x_j \right).\end{aligned}$$

Thus one large coupled problem is replaced by N independent subproblems:

$$\min_{x_j \in X_j} \left(f_j(x_j) - u^{\mathsf{T}} A_j x_j \right), \qquad j = 1, \ldots, N.$$

The subproblems may be solved independently and in parallel.

The multiplier coordinates otherwise independent decisions.

A separable quadratic example

Consider

$$\begin{aligned} \text{minimize} \quad & \tfrac{1}{2}\sum_{i=1}^{n} x_i^2 \\ \text{subject to} \quad & b - \sum_{i=1}^{n} a_i x_i \le 0, \\ & x_i \ge 0, \quad i = 1, \dots, n, \end{aligned}$$

where $a_i > 0$ and $b > 0$.

For a fixed multiplier $y \ge 0$, $L(x, y) = \dfrac{1}{2}\sum_{i=1}^{n} x_i^2 + y\left(b - \sum_{i=1}^{n} a_i x_i\right)$.

The relaxation separates into

$$\min_{x_i \ge 0} \left(\frac{1}{2}x_i^2 - y a_i x_i\right), \qquad i = 1, \dots, n.$$

Each scalar subproblem can be solved analytically.

Solving the quadratic relaxation

For each i, stationarity gives

$$x_i - y a_i = 0.$$

Since $y \geq 0$ and $a_i > 0$, the solution satisfies $x_i \geq 0$, and hence

$$\boxed{x_i(y) = y a_i.}$$

Substitution gives

$$\varphi(y) = L(x(y), y) = yb - \frac{1}{2} y^2 \sum_{i=1}^{n} a_i^2.$$

The dual problem is therefore

$$\underset{y \geq 0}{\text{maximize}} \quad yb - \frac{1}{2} y^2 \sum_{i=1}^{n} a_i^2.$$

Its solution is $\hat{y} = \dfrac{b}{\sum_{i=1}^{n} a_i^2}$.

Recovering the primal solution

Using

$$\hat{x}_i = \hat{y} a_i,$$

we obtain

$$\boxed{\hat{x}_i = \frac{b a_i}{\sum_{j=1}^{n} a_j^2}, \qquad i = 1, \ldots, n.}$$

The coupling constraint is active:

$$\sum_{i=1}^{n} a_i \hat{x}_i = b.$$

Moreover, $L(\hat{x}, \hat{y}) = f(\hat{x})$, so the primal and dual objective values agree.

Thus the relaxed subproblems, the dual multiplier, and complementarity recover and certify the global solution.

A coupled problem has been solved through independent scalar problems.

Multipliers as coordinating prices

In the separable problem,

$$f_j(x_j) - u^{\mathsf{T}} A_j x_j$$

is the local objective faced by subsystem j.

The multiplier u acts as a price for the shared resource or coupling constraint.

For a given price:

- each subsystem chooses its own optimal decision $x_j(u)$;
- the decisions can be computed independently;
- the resulting aggregate use is compared with the required balance.

The dual problem adjusts the price until the independent decisions are coordinated.

Dual variables provide both optimality information and coordination signals.

A glimpse of dual algorithms

For given multipliers $(u^{(k)}, y^{(k)})$, solve

$$x^{(k)} \in \operatorname*{argmin}_{x \in X} L(x, u^{(k)}, y^{(k)}).$$

The constraint residuals indicate how the multipliers may be changed:

$$u^{(k+1)} = u^{(k)} + \alpha_k h(x^{(k)}),$$

$$y^{(k+1)} = \left[y^{(k)} + \alpha_k g(x^{(k)}) \right]_+ .$$

Interpretation:

- increase the price of a violated constraint;
- decrease the price when the constraint has surplus;
- project inequality multipliers onto the nonnegative orthant.

A recurring strategy in optimization

Several methods in the course follow the same general pattern:

Build a simpler problem, solve it, and use its solution to update.

- The simplex method moves from one basic problem to the next.
- Newton's method minimizes local quadratic models.
- Gauss–Newton solves linear least-squares subproblems.
- Lagrangian relaxation removes coupling constraints and solves simpler subproblems.

The simpler problem should retain the information that matters while exposing structure that can be exploited computationally.

Solutions and certificates

Finding a candidate solution is only part of optimization. A certificate states what has been verified about the candidate.

- **LP:** primal feasibility, dual feasibility, and complementarity certify global optimality.
- **General nonlinear optimization:** KKT multipliers certify first-order stationarity, but not necessarily optimality.
- **Convex optimization:** KKT conditions certify global optimality.
- **Lagrangian duality:** equal primal and dual values certify global optimality.
- **Global optimality conditions:** global minimization of the Lagrangian, feasibility, and complementarity certify global optimality.

The meaning of a certificate depends on the assumptions and on the conclusion being certified.

Local models and global information

Local information

- gradients and Hessians;
- feasible and descent directions;
- Taylor models;
- Newton and Gauss–Newton steps;
- KKT conditions.

Global information

- convexity;
- primal and dual bounds;
- strong duality;
- complementarity;
- global optimality certificates.

Convexity turns local optimality conditions into global certificates.

Recognize and exploit structure

Different structures lead to different methods:

- **Linearity:** simplex methods, duality, and interior-point methods.
- **Network structure:** specialized flow algorithms.
- **Quadratic structure:** linear optimality systems and matrix factorizations.
- **Least-squares structure:** projections and Gauss–Newton.
- **Separability:** decomposition and parallel subproblems.
- **Convexity:** global conclusions from local conditions.

Do not treat an optimization problem as a black box.

Where does optimization go from here?

Large-scale linear and discrete optimization

- interior-point methods;
- column generation and decomposition;
- integer and combinatorial optimization.

Large-scale smooth optimization

- quasi-Newton and conjugate-gradient methods;
- line-search and trust-region methods.

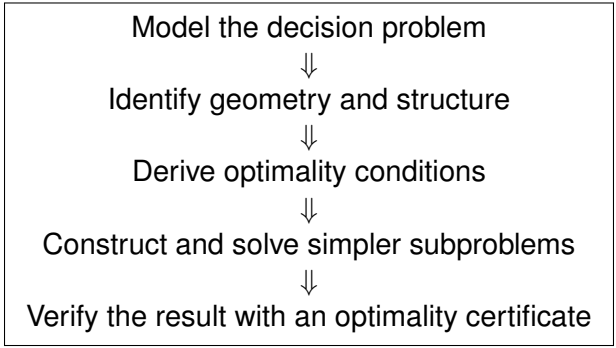
Constrained nonlinear optimization

- sequential quadratic programming;
- augmented Lagrangian and interior-point methods.

Data-driven optimization

- stochastic gradient methods;
- regularization and inverse problems;
- optimization in statistics and machine learning.

The course in one picture



The main recurring ideas are:

geometry, algebra, approximation, duality.

Main ideas

After this lecture, you should be able to:

- distinguish KKT stationarity from global minimization of the Lagrangian;
- state and use the global optimality conditions;
- explain why KKT conditions certify global optimality in convex problems;
- define the Lagrangian dual function and explain its lower-bound property;
- explain how relaxation of a coupling constraint can create independent subproblems;
- interpret dual variables as coordinating prices;
- identify the recurring roles of structure, local models, dual bounds, and optimality certificates throughout the course.

Optimization is the art of recognizing structure, constructing useful models, and obtaining both solutions and certificates.